

Pacific Refinery Company-

Production Optimisation

Using Linear Programming

(with LINGO)

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Q1

The objective is to determine the optimal combination of volumes of feedstocks and alcohols, to produce the required volumes of gasoline at the lowest cost. Pacific Refinery Company (PR co) produces regular, high-test and premium gasoline, by blending varying amounts of 89-octane and 95-octane feedstocks, with varying amounts of ethyl and methyl alcohols. Each type of gasoline produced must adhere to specified octane rating requirements and production targets, and there are specified amounts of feedstock and alcohols available.

Variables

Each variable represents the separate volumes (*in gallons*) of each type of feedstock (89-octane = F_{89x} , 95-octane = F_{95x}) and alcohol (Ethyl = E_x , Methyl = M_x) added to each type of gasoline blend (Regular = X_R , High-test = X_H , Premium = X_P). Note- Premium gasoline may only contain Ethyl alcohol, therefore there is no variable ' M_P '.

- F_{89R} ; gallons of 89-octane Feedstock added to Regular gasoline
- F_{89H} ; gallons of 89-octane Feedstock added to High-test gasoline
- F_{89P} ; gallons of 89-octane Feedstock added to Premium gasoline
- F_{95R} ; gallons of 95-octane Feedstock added to Regular gasoline
- F_{95H} ; gallons of 95-octane Feedstock added to High-test gasoline
- F_{95P} ; gallons of 95-octane Feedstock added to Premium gasoline
- E_R ; gallons of Ethyl alcohol added to Regular gasoline
- E_H ; gallons of Ethyl alcohol added to High-test gasoline
- E_P ; gallons of Ethyl alcohol added to Premium gasoline
- M_R ; gallons of Methyl alcohol added to Regular gasoline
- M_H ; gallons of Methyl alcohol added to High-test gasoline

The following variables can be added together to calculate the total volumes of each ingredient used, and gasoline produced:

- $F_{89R} + F_{89H} + F_{89P}$ = total volume of 89-octane Feedstock used
- $F_{95R} + F_{95H} + F_{95P}$ = total volume of 95-octane Feedstock used
- $E_R + E_H + E_P$ = total volume of Ethyl alcohol used
- $M_R + M_H + M_P$ = total volume of Methyl alcohol used

- $F_{89R} + F_{95R} + E_R + M_R$ = total volume of Regular gasoline produced
- $F_{89H} + F_{95H} + E_H + M_H$ = total volume of High-test gasoline produced
- $F_{89P} + F_{95P} + E_P$ = total volume of Premium gasoline produced

Objective function

The objective is to minimise the total production cost (C). This cost is calculated based on the total volumes of each ingredient used in gallons, multiplied by their respective prices per gallon (*in £s*). The objective function to be minimised is:

$$C = 1.00 \times (F89_R + F89_H + F89_P) + 1.50 \times (F95_R + F95_H + F95_P) + 1.25 \times (E_R + E_H + E_P) + 0.80 \times (M_R + M_H)$$

Constraints

The constraints represent the limitations, specifications and requirements of the variables in the problem.

Ingredient availability- There are limited amounts of feedstock and alcohols available (*in gallons*), so the following variable totals cannot exceed these amounts. These constraints represent the upper bound of these variables' domains:

- $F89_R + F89_H + F89_P \leq 18,000$ (89-octane Feedstock availability)
- $F95_R + F95_H + F95_P \leq 15,000$ (95-octane Feedstock availability)
- $E_R + E_H + E_P \leq 4000$ (Ethyl alcohol availability)
- $M_R + M_H \leq 5210$ (Methyl alcohol availability)

Production requirements- These are the required amounts of each type of gasoline which must be produced (*in gallons*), which means the totals of these variables must assumably be exactly equal to these amounts:

- $F89_R + F95_R + E_R + M_R = 15,000$ (Regular gasoline)
- $F89_H + F95_H + E_H + M_H = 12,000$ (High-test gasoline)
- $F89_P + F95_P + E_P = 6780$ (Premium gasoline)

Alcohol volume limits- These constraints limit the amounts of alcohols contained in the gasoline to specified percentages. Regular gasoline can contain up to 5% total alcohols; high-test up to 10% total alcohols; premium can only contain ethyl alcohol up to 5%. These constraints work because if the amounts of alcohol(s), $E + M$, is greater than the specified percentage of the total gasoline volume, $x\% \times (F89 + F95 + E + M)$, the left-hand-side would be positive, violating the constraint:

- $E_R + M_R - (0.05 \times (F89_R + F95_R + E_R + M_R)) \leq 0$ (Alcohols in Regular gasoline)
- $E_H + M_H - (0.10 \times (F89_H + F95_H + E_H + M_H)) \leq 0$ (alcohols in High-test gasoline)
- $E_P - (0.05 \times (F89_P + F95_P + E_P)) \leq 0$ (Ethyl alcohol in Premium gasoline)

Octane rating requirements- The average octane ratings for each type of gasoline must be at least 91 for regular gasoline, at least 93 for high-test, and at least 95 for premium. Octane ratings are calculated as the average octane ratings for the ingredients, weighted by volume. These constraints ensure that the total octane contribution from each ingredient, $89 \times F89 + 95 \times F95 + 120 \times E + 110 \times M$, is more than or equal to the total required octane, $x \times (F89 + F95 + E + M)$, otherwise the right-hand-side would be less than 0:

- $(89 \cdot F_{89R} + 95 \cdot F_{95R} + 120 \cdot E_R + 110 \cdot M_R) - (91 \cdot (F_{89R} + F_{95R} + E_R + M_R)) \geq 0$ (Regular gasoline)
- $(89 \cdot F_{89H} + 95 \cdot F_{95H} + 120 \cdot E_H + 110 \cdot M_H) - (93 \cdot (F_{89H} + F_{95H} + E_H + M_H)) \geq 0$ (High-test gasoline)
- $(89 \cdot F_{89P} + 95 \cdot F_{95P} + 120 \cdot E_P) - (95 \cdot (F_{89P} + F_{95P} + E_P)) \geq 0$ (Premium gasoline)

Non-negativity constraints- These make sure that none of the variable values fall below 0 to ensure a feasible, realistic solution. They also represent the lower bound of the variable domains:

- $F_{89R} \geq 0$
- $F_{89H} \geq 0$
- $F_{89P} \geq 0$
- $F_{95R} \geq 0$
- $F_{95H} \geq 0$
- $F_{95P} \geq 0$
- $E_R \geq 0$
- $E_H \geq 0$
- $E_P \geq 0$
- $M_R \geq 0$
- $M_H \geq 0$

Q2

Formulating and solving this problem as a linear programming (LP) model using LINGO, provides the following relevant results (full output can be found in the appendix):

Global optimal solution found.

Objective value:

40220.25

Variable	Value	Reduced Cost
F89R	9587.500	0.000000
F89H	7000.000	0.000000
F89P	1412.500	0.000000
F95R	4662.500	0.000000
F95H	3800.000	0.000000
F95P	5028.500	0.000000
ER	0.000000	0.4500000
EH	0.000000	0.4500000
EP	339.0000	0.000000
MR	750.0000	0.000000
MH	1200.000	0.000000

From these results, an optimal production plan can be created- these are the optimum volumes of each ingredient to be blended, to produce each type of gasoline.

Optimal production plan:

1. Regular gasoline:

F_{89R} = 9587.5 gallons of 89-Octane feedstock, +
 F_{95R} = 4662.5 gallons of 95-octane feedstock, +
 E_R = 0 gallons of Ethyl alcohol, +
 M_R = 750 gallons of Methyl alcohol

2. High-test gasoline:

F_{89H} = 7000 gallons of 89-octane feedstock, +
 F_{95H} = 3800 gallons of 95-octane feedstock, +
 E_H = 0 gallons of Ethyl alcohol, +
 M_H = 1200 gallons of Methyl alcohol

3. Premium gasoline:

F_{89P} = 1412.5 gallons of 89-octane feedstock, +
 F_{95P} = 5028.5 gallons of 95-octane feedstock, +
 E_P = 339 gallons of Ethyl alcohol

Optimal production plan cost:

The LP provides an objective value of 40220.25- therefore, the final cost of this production plan would be **£40,220.25**. This is calculated using the objective function:

$$C = 1.00 * (F_{89R} + F_{89H} + F_{89P}) + 1.50 * (F_{95R} + F_{95H} + F_{95P}) + 1.25 * (E_R + E_H + E_P) + 0.80 * (M_R + M_H) =$$

$$1.00 * (11875 + 4712.5 + 1412.5) + 1.50 * (2375 + 6087.5 + 5028.5) + 1.25 * (0 + 0 + 339) + 0.80 * (750 + 1200) =$$

40,220.25

Alternative optimal solution

Currently, the gasoline production constraint assumes that production of each type of gasoline must be *exactly equal* to 15,000 gallons for Regular, 12,000 gallons for High-test and 6780 gallons for Premium. An alternative to this would be to simply change the (=) constraint to ($\geq$), making the constraints less strict. This would mean the production requirements would still be at least met, while allowing some flexibility for excess gasoline to be potentially produced, which could be passed onto the next day for example if necessary:

- $F_{89R} + F_{95R} + E_R + M_R \geq 15,000$ (Regular gasoline)
- $F_{89H} + F_{95H} + E_H + M_H \geq 12,000$ (High-test gasoline)
- $F_{89P} + F_{95P} + E_P \geq 6780$ (Premium gasoline)

This adjustment to the model provides the following relevant results (full output and LINGO formulation for alternative LP can be found in the appendix):

Global optimal solution found.
Objective value:

40220.25

Variable	Value	Reduced Cost
F89R	11000.00	0.000000
F89H	7000.000	0.000000
F89P	0.000000	0.000000
F95R	3250.000	0.000000
F95H	3800.000	0.000000
F95P	6441.000	0.000000
ER	0.000000	0.4500000
EH	0.000000	0.4500000
EP	339.0000	0.000000
MR	750.0000	0.000000
MH	1200.000	0.000000

From these results, an alternative optimal production plan can be created - these are the optimum volumes of each ingredient to be blended to produce each type of gasoline:

1. Regular Gasoline:

F89_R = 11000 gallons of 89-octane feedstock, +
F95_R = 3250 gallons of 95-octane feedstock, +
E_R = 0 gallons of Ethyl alcohol, +
M_R = 750 gallons of Methyl alcohol

2. High-Test Gasoline:

F89_H = 7000 gallons of 89-octane feedstock
F95_H = 3800 gallons of 95-octane feedstock
E_H = 0 gallons of Ethyl alcohol
M_H = 1200 gallons of Methyl alcohol

3. Premium Gasoline:

F89_P = 0 gallons of 89-octane feedstock
F95_P = 6441 gallons of 95-octane feedstock
E_P = 339 gallons of Ethyl alcohol

The alternative LP also provides an objective value of 40220.25- therefore, the final cost of this production plan would also be **£40,220.25**.

This shows that the same amounts of gasoline will be produced at the same cost, while still complying with all the requirements, however the quantities of some ingredients added to the gasolines vary. This gives PR co some flexibility with ingredient usage, without having to worry about production volumes, or price changes, or gasoline not meeting alcohol and octane requirements. For example, if there is an unexpected issue with 89-octane feedstock availability, they could safely choose the alternative plan, which doesn't require any of this feedstock for Premium gasoline.

Below is a direct comparison of the original vs. alternative production plans for each gasoline type:

	Original production plan volumes (in gallons)	Alternative production plan volumes (in gallons)
89-octane feedstock	9587.5	11,000
95-octane feedstock	4662.5	3250
Ethyl alcohol	0	0
Methyl alcohol	750	750
Regular gasoline total	15,000	15,000
89-octane feedstock	7000	7000
95-octane feedstock	3800	3800
Ethyl alcohol	0	0
Methyl alcohol	1200	1200
High-test gasoline total	12,000	12,000
89-octane feedstock	1412.5	0
95-octane feedstock	5028.5	6441
Ethyl alcohol	339	339
Premium gasoline total	6780	6780

Q3

i)

To analyse this problem, the results of two separate variations of the LP will be compared to the original model, to evaluate which scenario would be most optimal. The original model assumes BC co's offer is not accepted, and the two variations will assume that the offer is accepted, with either Ethyl (variation 1) or Methyl (variation 2) being increased in each option.

In each variation, firstly, the ingredient availability constraints of Ethyl or Methyl respectively will be increased by 300:

- $E_R + E_H + E_P \leq 4300$ (Ethyl alcohol availability) (variation 1)
- $M_R + M_H \leq 5510$ (Methyl alcohol availability) (variation 2)

Secondly, the objective function of each will have to be adjusted to accommodate the change in price of each alcohol, as the objective function contains the costs per gallon of the ingredients.

The current average cost of Ethyl is £1.25 per gallon, at 4000 gallons. By adding 300 gallons at £0.30 per gallon, there will now be 4300 gallons, with a new average cost instead of £1.25:

$$\text{Current total cost} = 4000 \text{ gallons} * £1.25 = £5000$$

*Additional Ethyl cost = 300 gallons * £0.30 = £90*

New total cost = 5000 + 90 = £5090

New total volume = 4000 + 300 = 4300 gallons

New average cost = 5090/4300 ≈ £1.18...

For accuracy, (5090/4300) will be used instead of £1.18.

Therefore, the new objective function for variation 1 will be:

$$C = 1.00 \times (F89_R + F89_H + F89_P) + 1.50 \times (F95_R + F95_H + F95_P) + (5090/4300) \times (E_R + E_H + E_P) + 0.80 \times (M_R + M_H)$$

Repeating the same calculations for variation 2 with Methyl, means the new average cost of Methyl per gallon will be 4258/5510 ($\approx$ £0.77...), instead of £0.80.

Therefore, the new objective function for variation 2 will be:

$$C = 1.00 \times (F89_R + F89_H + F89_P) + 1.50 \times (F95_R + F95_H + F95_P) + 1.25 \times (E_R + E_H + E_P) + (4258/5510) \times (M_R + M_H)$$

Comparison of results

After implementing these changes, below is a comparison of the outputs of the original LP, with the two variations where additional Ethyl is accepted, and where additional Methyl is accepted (full outputs of both variations can be found in the appendix). This table represents the total plan costs, how much of each ingredient is used (per type of gasoline, and in total) (full outputs and LINGO formulations of each variation can be found in the appendix):

	Original	Variation 1 (Ethyl)	Variation 2 (Methyl)
Objective value	40,220.25	40197.78	40167.16
F89 _R	9587.5	9587.5	9587.5
F89 _H	7000	7000	7000
F89 _P	1412.5	1412.5	1412.5
Total F89	18,000	18,000	18,000
F95 _R	4662.5	4662.5	4662.5
F95 _H	3800	3800	3800
F95 _P	5028.5	5028.5	5028.5
Total F95	13,491	13,491	13,491
E _R	0	0	0
E _H	0	0	0
E _P	339	339	339
Total Ethyl	339	339	339
M _R	750	750	750
M _H	1200	1200	1200
Total Methyl	1950	1950	1950

According to these results, PR co should **accept** BC co's offer, and take the extra 300 gallons of **Methyl alcohol**. This will result in a minor saving in the overall cost of £53.09, compared to a smaller £22.47 saving with the Ethyl offer. Neither change has any effect on the optimum plan - the exact same quantities of each ingredient will be used for each type of gasoline. This is likely because relatively very low amounts of the alcohols are used in the original plan anyway, and the reduction in overall cost likely just represents the decrease in average price of each alcohol, multiplied by the volume of the alcohol used; since more methyl is used (1950 gallons) than ethyl (339 gallons), this results in a greater reduction in total cost for the Methyl offer.

ii)

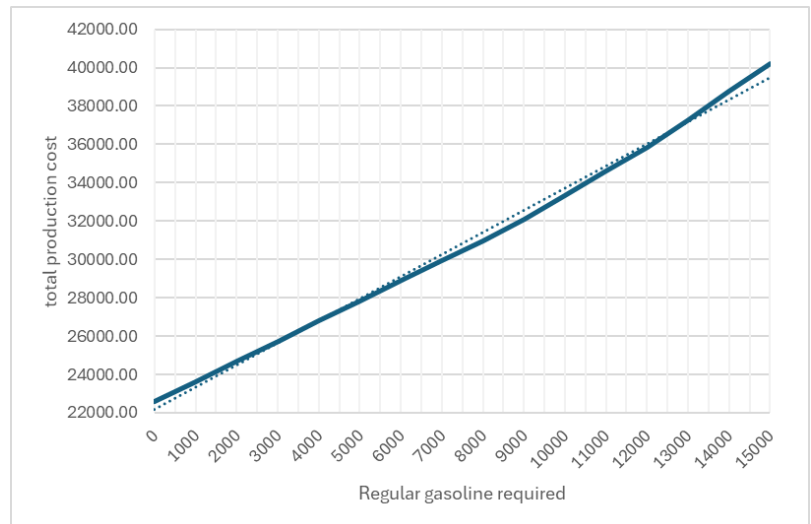
To evaluate the general relationship between the total final production cost and the amount of required regular gasoline, the original LP must be run multiple times, decreasing the gasoline production constraint for regular gasoline each time by 1000 while leaving the rest of the program the same:

- Run 0- $F_{89R} + F_{95R} + E_R + M_R = 15000$;
- Run 1- $F_{89R} + F_{95R} + E_R + M_R = 14000$;
- Run 2- $F_{89R} + F_{95R} + E_R + M_R = 13000$...

Doing this until the constraint = 0, provides the following results. The change in total production cost with every run is also recorded:

Run number	Regular gasoline required (gallons)	Total production cost (£)	Change in total production cost (£)
0	15000	40220.25	-
1	14000	38755.25	-1465.00
2	13000	37290.25	-1465.00
3	12000	35848.88	-1441.37
4	11000	34597.62	-1251.26
5	10000	33346.38	-1251.24
6	9000	32095.12	-1251.26
7	8000	30979.00	-1116.12
8	7000	29929.00	-1050.00
9	6000	28879.00	-1050.00
10	5000	27829.00	-1050.00
11	4000	26779.00	-1050.00
12	3000	25729.00	-1050.00
13	2000	24679.00	-1050.00
14	1000	23629.00	-1050.00
15	0	22579.00	-1050.00

This graph visually shows the above data- the relationship between the amount of required gasoline and the total cost. The correlation looks nearly linear- the straight dotted trendline is there as a comparison to show that it is not quite linear. However, from this graph it is difficult to specify the level of nonlinearity. The 'change in total production cost' column quantifies this, and shows that the correlation is in fact *partially* linear. From 0 to between 7000/8000 required gallons, the total cost steadily increases by £1050 every 1000 gallons. After this, the relationship complicates. The gradient fluctuates slightly between around 8000-13,000 gallons, then seems to stabilise after 12-13,000 gallons.



Generally, if the production requirements were to decrease from 15,000, this would decrease the total production cost by £1.465* per every gallon decreased (£1465 / 1000 gallons), for at least the first 2000 gallons decreased. After this point, the rate at which total production cost decreases, decelerates with some fluctuation until around 7000 gallons. If the production requirements were to decrease below around 7000, then the total production cost would decrease nearly unitarily at £1.05** per gallon (£1050 / 1000 gallons).

If PR co needed more precise data on this, a more tedious, granular approach with smaller intervals (than 1000) of required gasoline would be required, to pinpoint exactly where and how the gradient fluctuates.

Using shadow/dual price

These results, particularly the 'change in total production cost' column, have a very significant relationship with the outputted 'dual price column' of this LP- specifically row 6, which corresponds to the production requirement constraint for regular gasoline:

...	Row	Slack or Surplus	Dual Price
	1	40220.25	-1.000000
	2	0.000000	0.500000
	3	1509.000	0.000000
	4	3661.000	0.000000
	5	3260.000	0.000000
	6	0.000000	-1.465000
	7	...	

This is the output at run 0- the dual price = -1.465*, a number mentioned previously. The dual, or shadow price, represents the rate of improvement in an objective function per unit increase in the right-hand-side of the constraint. Therefore, row 6 of this dual price column should provide the same results as the change in total production cost column (divided by 1000, as this column shows the change per 1000

change in the right-hand-side of the constraint, rather than per unit change)- and it does, as they are essentially showing the same thing on a different scale.

The results show that the linearity of the relationship between total cost and amount of regular gasoline needed, (/the gradient of the graph, or the 'constantness' of the dual price), seem to change at two points- around 12-13,000 gallons, and around 8000-9000 gallons. Using a potentially tedious, manual process of adjusting the RHS of the constraint at around these points (playing 'higher or lower' until the dual price value changes, e.g. 12,000, then 13,000, then 12,500, then 12,250 etc) and checking the output, will allow to pinpoint at which exact point the dual price value changes, and therefore provide a more accurate representation of the relationship between cost and amount of gasoline required.

Using this method shows that at $F89_R + F95_R + E_R + M_R = 12111$ the dual price is still at -1.465000 - however changes to -1.251250 at 12110 .

Again, $F89_R + F95_R + E_R + M_R = 8672$, the dual price is at -1.251250 , and changes to -1.050000 at 8671 .

Implications

This means that if the requirement for regular gasoline was to decrease from 15,000 by 2889 gallons, the decrease in total cost will be linear at £1.465 per gallon decreased. After this point, a further decrease in the requirement for regular gasoline from 12,110 to 8672 gallons, will result in a linear decrease in cost of £1.25125 per gallon. A decrease from 8671 to 0 gallons required, will result in a £1.05** decrease in total cost, per gallon.

This proves that the relationship is in a way linear, but with 2 'kinks' in linearity at 8672 and 12,111 gallons of required regular gasoline- the relationship is linear between 0 - 8671 gallons with a 'gradient' of 1.05; linear between 8672 - 12,110 gallons with a 'gradient' of 1.25125; and linear between 12,111-15,000+ gallons with a 'gradient' of 1.465.

Q4

i)

To ensure that PR uses at least twice as much Ethyl alcohol as Methyl alcohol, a new constraint must be added that represents this. The original LP formulation can stay the same (converted to standard form), with the exact same objective function, variables and constraints- the additional constraint to be implemented is:

$$E_R + E_H + E_P - 2 * (M_R + M_H) \geq 0$$

$E_R + E_H + E_P$ represents the total volume of Ethyl alcohol used throughout all the gasoline types, and $M_R + M_H$ represents the total volume of Methyl alcohol used. This

constraint ensures that the total Ethyl is equal to or greater than two times the total Methyl, otherwise the right-hand-side would be less than 0.

Converting into standard form so that it is an equality (and that the RHS is non-negative which it already is, as 0 is sufficient):

$$E_R + E_H + E_P - 2 * (M_R + M_H) + S_x = 0$$

(All other constraints converted to standard form should also contain slack variables S_2, S_3 etc- in S_x case, x may vary depending on the order of the constraints- e.g. if this is the first constraint in the LP, it would be S_1 so that each constraint has a separate slack variable.)

ii)

A new variable must be introduced, Y , that represents the shortfall, in gallons, of high-test gasoline production- this variable will be added to the production requirements constraint for high-test gasoline:

$$F89_H + F95_H + E_H + M_H + Y = 12000$$

This will mean that Y = the shortfall between what is required, minus how much would be produced- rearranging the equation, means that:

$$Y = 12000 - (F89_H + F95_H + E_H + M_H)$$

Converting the constraint to standard form by adding a slack variable:

$$F89_H + F95_H + E_H + M_H + Y + S_x = 12000$$

(All other constraints converted to standard form should also contain slack variables S_2, S_3 etc- in S_x case, x may vary depending on the order of the constraints- e.g. if this is the first constraint in the LP, it would be S_1 so that each constraint has a separate slack variable.)

Additionally, this constraint must be added to the objective function, as it is another variable that contributes to the total cost, with a 'price'/coefficient of £2 per gallon:

$$C = 1.00 \times (F89_R + F89_H + F89_P) + 1.50 \times (F95_R + F95_H + F95_P) + 1.25 \times (E_R + E_H + E_P) + 0.80 \times (M_R + M_H) + 2 * Y$$

Lastly, this new variable must also be non-negative:

$$Y \geq 0$$

Appendix

Q2 Original LP output:

LINGO/WIN64 21.0.33 (29 May 2024), LINDO API 15.0.6099.171

Licensee info: Eval Use Only
License expires: 18 JAN 2025

Global optimal solution found.

Objective value:	40220.25
Infeasibilities:	0.000000
Total solver iterations:	8
Elapsed runtime seconds:	0.13

Model Class:	LP
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Total variables:	11
Nonlinear variables:	0
Integer variables:	0

Total constraints:	25
Nonlinear constraints:	0

Total nonzeros:	65
Nonlinear nonzeros:	0

Variable	Value	Reduced Cost
F89R	9587.500	0.000000
F89H	7000.000	0.000000
F89P	1412.500	0.000000
F95R	4662.500	0.000000
F95H	3800.000	0.000000
F95P	5028.500	0.000000
ER	0.000000	0.4500000
EH	0.000000	0.4500000
EP	339.0000	0.000000
MR	750.0000	0.000000
MH	1200.000	0.000000

Row	Slack or Surplus	Dual Price
1	40220.25	-1.000000
2	0.000000	0.5000000
3	1509.000	0.000000
4	3661.000	0.000000
5	3260.000	0.000000
6	0.000000	-1.465000
7	0.000000	-1.430000
8	0.000000	-1.487500
9	0.000000	0.7000000
10	0.000000	0.7000000
11	0.000000	0.2500000
12	13725.00	0.000000
13	0.000000	0.000000
14	0.000000	0.000000
15	9587.500	0.000000
16	4662.500	0.000000
17	0.000000	0.000000
18	750.0000	0.000000
19	7000.000	0.000000
20	3800.000	0.000000
21	0.000000	0.000000
22	1200.000	0.000000
23	1412.500	0.000000
24	5028.500	0.000000
25	339.0000	0.000000

Q2 Adjusted/ Alternative LP output:

LINGO/WIN64 21.0.33 (29 May 2024), LINDO API 15.0.6099.171

Licensee info: Eval Use Only
License expires: 18 JAN 2025

Global optimal solution found.
Objective value: 40220.25
Infeasibilities: 0.000000
Total solver iterations: 9
Elapsed runtime seconds: 0.12

Model Class: LP

Total variables: 11
Nonlinear variables: 0
Integer variables: 0

Total constraints: 25
Nonlinear constraints: 0

Total nonzeros: 65
Nonlinear nonzeros: 0

Variable	Value	Reduced Cost
F89R	11000.00	0.000000
F89H	7000.000	0.000000
F89P	0.000000	0.000000
F95R	3250.000	0.000000
F95H	3800.000	0.000000
F95P	6441.000	0.000000
ER	0.000000	0.4500000
EH	0.000000	0.4500000
EP	339.0000	0.000000
MR	750.0000	0.000000
MH	1200.000	0.000000

Row	Slack or Surplus	Dual Price
1	40220.25	-1.000000
2	0.000000	0.5000000
3	1509.000	0.000000
4	3661.000	0.000000
5	3260.000	0.000000
6	0.000000	-1.465000
7	0.000000	-1.430000
8	0.000000	-1.487500
9	0.000000	0.7000000
10	0.000000	0.7000000
11	0.000000	0.2500000
12	5250.000	0.000000
13	0.000000	0.000000
14	8475.000	0.000000
15	11000.00	0.000000
16	3250.000	0.000000
17	0.000000	0.000000
18	750.0000	0.000000
19	7000.000	0.000000
20	3800.000	0.000000
21	0.000000	0.000000
22	1200.000	0.000000
23	0.000000	0.000000
24	6441.000	0.000000
25	339.0000	0.000000

Q3 Variation 1 (Ethyl) LP output:

LINGO/WIN64 21.0.33 (29 May 2024), LINDO API 15.0.6099.171

Licensee info: Eval Use Only
License expires: 18 JAN 2025

Global optimal solution found.
Objective value: 40197.78
Infeasibilities: 0.000000
Total solver iterations: 8
Elapsed runtime seconds: 0.13

Model Class: LP

Total variables: 11
Nonlinear variables: 0
Integer variables: 0

Total constraints: 25
Nonlinear constraints: 0

Total nonzeros: 65
Nonlinear nonzeros: 0

Variable	Value	Reduced Cost
F89R	9587.500	0.000000
F89H	7000.000	0.000000
F89P	1412.500	0.000000
F95R	4662.500	0.000000
F95H	3800.000	0.000000
F95P	5028.500	0.000000
ER	0.000000	0.3837209
EH	0.000000	0.3837209
EP	339.0000	0.000000
MR	750.0000	0.000000
MH	1200.000	0.000000

Row	Slack or Surplus	Dual Price
1	40197.78	-1.000000
2	0.000000	0.5000000
3	1509.000	0.000000
4	3961.000	0.000000
5	3260.000	0.000000
6	0.000000	-1.465000
7	0.000000	-1.430000
8	0.000000	-1.484186
9	0.000000	0.7000000
10	0.000000	0.7000000
11	0.000000	0.3162791
12	13725.00	0.000000
13	0.000000	0.000000
14	0.000000	0.000000
15	9587.500	0.000000
16	4662.500	0.000000
17	0.000000	0.000000
18	750.0000	0.000000
19	7000.000	0.000000
20	3800.000	0.000000
21	0.000000	0.000000
22	1200.000	0.000000
23	1412.500	0.000000
24	5028.500	0.000000
25	339.0000	0.000000

Q3 Variation 2 (Methyl) LP output:

LINGO/WIN64 21.0.33 (29 May 2024), LINDO API 15.0.6099.171

Licensee info: Eval Use Only
License expires: 18 JAN 2025

Global optimal solution found.
Objective value: 40167.16
Infeasibilities: 0.000000
Total solver iterations: 8
Elapsed runtime seconds: 0.13

Model Class: LP

Total variables: 11
Nonlinear variables: 0
Integer variables: 0

Total constraints: 25
Nonlinear constraints: 0

Total nonzeros: 65
Nonlinear nonzeros: 0

Variable	Value	Reduced Cost
F89R	9587.500	0.000000
F89H	7000.000	0.000000
F89P	1412.500	0.000000
F95R	4662.500	0.000000
F95H	3800.000	0.000000
F95P	5028.500	0.000000
ER	0.000000	0.4772232
EH	0.000000	0.4772232
EP	339.0000	0.000000
MR	750.0000	0.000000
MH	1200.000	0.000000

Row	Slack or Surplus	Dual Price
1	40167.16	-1.000000
2	0.000000	0.5000000
3	1509.000	0.000000
4	3661.000	0.000000
5	3560.000	0.000000
6	0.000000	-1.463639
7	0.000000	-1.427278
8	0.000000	-1.487500
9	0.000000	0.7272232
10	0.000000	0.7272232
11	0.000000	0.2500000
12	13725.00	0.000000
13	0.000000	0.000000
14	0.000000	0.000000
15	9587.500	0.000000
16	4662.500	0.000000
17	0.000000	0.000000
18	750.0000	0.000000
19	7000.000	0.000000
20	3800.000	0.000000
21	0.000000	0.000000
22	1200.000	0.000000
23	1412.500	0.000000
24	5028.500	0.000000
25	339.0000	0.000000

LINGO LP formulations

I've attached the LINGO code for the alternative of Q2 and both variations of Q3i here in case they were needed, as there is a limit of 3 files to the coursework submission-

Q2 Alternative

```
! Q2 Alternative/Adjusted LP

! Objective Function: Minimize total cost;
min = 1.00 * (F89R + F89H + F89P) + 1.50 * (F95R + F95H + F95P) + 1.25 * (ER + EH + EP) + 0.80 * (MR + MH);

! Subject to Constraints

! Ingredient availability;
F89R + F89H + F89P <= 18000;
F95R + F95H + F95P <= 15000;
ER + EH + EP <= 4000;
MR + MH <= 5210;

! Gasoline production requirements;
F89R + F95R + ER + MR >= 15000; ! >=;
F89H + F95H + EH + MH >= 12000; ! >=;
F89P + F95P + EP >= 6780; ! >=;

! Alcohol percentage volume limits;
ER + MR - (0.05 * (F89R + F95R + ER + MR)) <= 0 ;
EH + MH - (0.10 * (F89H + F95H + EH + MH)) <= 0 ;
EP - (0.05 * (F89P + F95P + EP)) <= 0;

! Octane rating requirements;
(89*F89R + 95*F95R + 120*ER + 110*MR) - (91 * (F89R + F95R + ER + MR)) >= 0 ;
(89*F89H + 95*F95H + 120*EH + 110*MH) - (93 * (F89H + F95H + EH + MH)) >= 0 ;
(89*F89P + 95*F95P + 120*EP) - (95 * (F89P + F95P + EP)) >= 0;

! Non-negativity constraints;
F89R >= 0; F95R >= 0; ER >= 0; MR >= 0;
F89H >= 0; F95H >= 0; EH >= 0; MH >= 0;
F89P >= 0; F95P >= 0; EP >= 0;

end
```

Q3i Variation 1- Ethyl

```
! Q3 i) Variation 1 (Additional Ethyl)

! Objective Function: Minimize total cost- adjusted cost of Ethyl;
min = 1.00 * (F89R + F89H + F89P) + 1.50 * (F95R + F95H + F95P) +
(5090/4300) * (ER + EH + EP) + 0.80 * (MR + MH);

! Subject to Constraints

! Ingredient availability;
F89R + F89H + F89P <= 18000;
F95R + F95H + F95P <= 15000;
ER + EH + EP <= 4300; ! +300 Methyl;
MR + MH <= 5210;

! Gasoline production requirements;
F89R + F95R + ER + MR = 15000;
F89H + F95H + EH + MH = 12000;
F89P + F95P + EP = 6780;

! Alcohol percentage volume limits;
```

```

ER + MR - (0.05 * (F89R + F95R + ER + MR)) <= 0 ;
EH + MH - (0.10 * (F89H + F95H + EH + MH)) <= 0 ;
EP - (0.05 * (F89P + F95P + EP)) <= 0;

! Octane rating requirements;
(89*F89R + 95*F95R + 120*ER + 110*MR) - (91 * (F89R + F95R + ER + MR)) >= 0 ;
(89*F89H + 95*F95H + 120*EH + 110*MH) - (93 * (F89H + F95H + EH + MH)) >= 0 ;
(89*F89P + 95*F95P + 120*EP) - (95 * (F89P + F95P + EP)) >= 0;

! Non-negativity constraints;
F89R >= 0; F95R >= 0; ER >= 0; MR >= 0;
F89H >= 0; F95H >= 0; EH >= 0; MH >= 0;
F89P >= 0; F95P >= 0; EP >= 0;

end

```

Q3i variation 2- Methyl

```

! Q3 i) Variation 2 (Additional Methyl)

! Objective Function: Minimize total cost- adjusted cost of Methyl;
min = 1.00 * (F89R + F89H + F89P) + 1.50 * (F95R + F95H + F95P) +
1.25 * (ER + EH + EP) + (4258/5510) * (MR + MH);

! Subject to Constraints

! Ingredient availability;
F89R + F89H + F89P <= 18000;
F95R + F95H + F95P <= 15000;
ER + EH + EP <= 4000;
MR + MH <= 5510; ! +300 Methyl

! Gasoline production requirements;
F89R + F95R + ER + MR = 15000;
F89H + F95H + EH + MH = 12000;
F89P + F95P + EP = 6780;

! Alcohol percentage volume limits;
ER + MR - (0.05 * (F89R + F95R + ER + MR)) <= 0 ;
EH + MH - (0.10 * (F89H + F95H + EH + MH)) <= 0 ;
EP - (0.05 * (F89P + F95P + EP)) <= 0;

! Octane rating requirements;
(89*F89R + 95*F95R + 120*ER + 110*MR) - (91 * (F89R + F95R + ER + MR)) >= 0 ;
(89*F89H + 95*F95H + 120*EH + 110*MH) - (93 * (F89H + F95H + EH + MH)) >= 0 ;
(89*F89P + 95*F95P + 120*EP) - (95 * (F89P + F95P + EP)) >= 0;

! Non-negativity constraints;
F89R >= 0; F95R >= 0; ER >= 0; MR >= 0;
F89H >= 0; F95H >= 0; EH >= 0; MH >= 0;
F89P >= 0; F95P >= 0; EP >= 0;

end

```